FLIGHT CONTROLLER

ARKV6X

Triple-IMU FMUv6X autopilot module on the Pixhawk Autopilot Bus

The USA-built ARKV6X flight controller is based on the FMUv6X and Pixhawk Autopilot Bus (PAB) open source standards. An STM32H743 processor reads three time-synchronised IMUs, each on its own SPI bus and switched 3.3 V rail, for redundant inertial sensing.

It fits any Pixhawk Autopilot Bus carrier, including the ARK PAB Carrier, ARK Jetson PAB Carrier and ARK Jetson PAB V3.



KEY FEATURES

- Three synchronised IMUs, each independently powered
  - Standard: 2 × TDK InvenSense ICM-42688-P (SPI2, SPI3) and 1 × IIM-42652 industrial IMU (SPI1)
  - Extended Range: 3 × TDK InvenSense IIM-42653 industrial IMU (SPI1–3), ±32 g and ±4000 °/s
- Bosch BMP390 barometer and BMM150 magnetometer on internal I2C4
- Four independently switched 3.3 V sensor rails — one per sensor domain
- Cypress FM25V02A 256 kbit FRAM for non-volatile parameter storage
- 1 W sensor heater under closed-loop control
- Supercapacitor-backed real-time clock
- microSD logging on a 4-bit SDMMC interface

SPECIFICATIONS

PROCESSOR & MEMORY	
MCU	ST STM32H743IIK6, 176-UFBGA
Flash / RAM	2 MB / 1 MB
FRAM	FM25V02A, 256 kbit, 40 MHz SPI
Oscillators	16.000 MHz + 32.768 kHz ±20 ppm

ELECTRICAL	
Supply voltage	4.8 V minimum, 5.5 V absolute maximum
Supply current	500 mA max at 5 V
— main system	300 mA
— sensor heater	200 mA
Operating temperature	−10 °C to +60 °C
Storage temperature	−10 °C to +60 °C

MECHANICAL	
Dimensions	36 × 29 × 5 mm
Weight	5.0 g
Mounting	4 × Ø2.3 mm on 31.4 × 24.4 mm

AT A GLANCE	
Standard	FMUv6X / PAB
MCU	STM32H743IIK6
IMUs	3, synchronised
Baro / Mag	BMP390 / BMM150
Supply	4.8–5.5 V, 500 mA
Operating temp	−10 to +60 °C
Interface	100 + 50 pin PAB
Dimensions	36 × 29 × 5 mm
Weight	5.0 g
Hardware rev	Rev 4.1 / 5.1

COMPLIANCE & SOURCING

- Designed and manufactured in the USA
- NDAA compliant
- Blue UAS Framework listed
- FMUv6X open standard
- Pixhawk Autopilot Bus DS-010 standard

ORDERING & RESOURCES

STORE arkelectron.com/product/arkv6x/
arkelectron.com/product/arkv6x-iim-42653/

DOCS docs.arkelectron.com/products/flight-controller/arkv6x

CAD github.com/ARK-Electronics/ark-hardware

VARIANTS

	ARKV6X	ARKV6X EXTENDED RANGE
IMU on SPI1	TDK InvenSense IIM-42652	TDK InvenSense IIM-42653
IMUs on SPI2, SPI3	TDK InvenSense ICM-42688-P	TDK InvenSense IIM-42653
Accel full scale	±16 g	±32 g
Gyro full scale	±2000 °/s	±4000 °/s
Hardware revision ID	ARKV6X000	ARKV6X001
Schematic	Rev 4.1	Rev 5.1

Same board, connectors and pinout.

PIXHAWK AUTOPILOT BUS INTERFACE

PAB CONNECTOR	HIROSE PART	SIGNALS CARRIED
PAB X1 (J2)	DF40C-100DP-0.4V, 100 pos	8 × PWM (DShot on 1–6) · CAN1 / CAN2 · 8 × UART (see serial map) · I2C1–I2C3 · USB with VBUS sense · PPM and RC input · SWD · safety switch, buzzer, armed · 3 × power-input sense · ADC · 5 V input
PAB X2 (J3)	DF40C-50DP-0.4V, 50 pos	Ethernet RMII with power enable · external SPI6 (2 × CS, reset, 2 × DRDY) · SPIX_SYNC · 17 reserved pins

SENSOR BUS MAP

DEVICE	INTERFACE	ADDRESS / SELECT	FUNCTION
IIM-42652 or IIM-42653 (U9)	SPI1	nCS1	Industrial 6-axis accel / gyro
ICM-42688-P or IIM-42653 (U12)	SPI2	nCS1	6-axis accelerometer / gyroscope
ICM-42688-P or IIM-42653 (U13)	SPI3	nCS1	6-axis accelerometer / gyroscope
BMP390 (U11)	I2C4	0x76	Barometric pressure
BMM150 (U10)	I2C4	0x10	3-axis magnetometer

Where two IMUs are named, the first is on the ARKV6X and the second on the Extended Range.

CARRIER BOARD OPTIONS

ARK CARRIER BOARD	AVIONICS I/O	ADDS
ARK Pixhawk Autopilot Bus Carrier	JST-GH	Full Pixhawk-standard breakout, 100 Mbps Ethernet, dual power
ARK Jetson PAB Carrier	JST-GH	NVIDIA Jetson Orin compute on Pixhawk-standard ports
ARK Jetson PAB V3	Pico-Clasp	Jetson Orin compute, triple power inputs, Ethernet switch
ARK VOXL2 RTK PAB Carrier	Pico-Clasp	ModalAI VOXL2 compute; Blue UAS Framework listed

SERIAL REFERENCE & FIRMWARE SUPPORT

SERIAL PORT MAPPING

UART	DEVICE	PORT
USART1	/dev/ttyS0	GPS
USART2	/dev/ttyS1	TELEM3
USART3	/dev/ttyS2	Debug Console
UART4	/dev/ttyS3	UART4 & I2C
UART5	/dev/ttyS4	TELEM2
USART6	/dev/ttyS5	PX4IO/RC
UART7	/dev/ttyS6	TELEM1
UART8	/dev/ttyS7	GPS2

FIRMWARE & SOFTWARE SUPPORT

- PX4 Autopilot — ships by default; ArduPilot supported
- Requires a PAB carrier: all I/O is on the two board-to-board connectors.
- Includes a microSD card.

ARK-DS-ARKV6X | Version 1.0, 2026-10-29 | Compiled from the ARKV6X Production schematics Rev 4.1 (13 Dec 2023) and Rev 5.1 Extended Range (19 Jul 2024), the ARKV6X Production bill of materials, the ARKV6X production STEP model, and ARK Electronics published documentation. Temperature ratings per component data sheets. Specifications are subject to change without notice. © 2026 ARK Electronics LLC.